Session 3 - Research Protocol v0.1

1. Title

A fragmentation-prediction prototype for economic assessment of blast-design decisions in sublevel stoping at an underground mine in southern Peru.

2. Abstract

Not developed yet, following the recorded oral instruction. The PPT requests ten sections, so the heading is retained to make the discrepancy visible.

3. Introduction and problem

High explosive consumption, oversize fragments, transport-budget deviations and missed monthly tonnage targets are reported, not yet quantified. The existing standard design prepared using AEGIS and technical calculations is the operational comparator.

4. Literature review

Systematic synthesis remains pending, consistent with the oral instruction not to develop it yet. Session 4 contains real candidate sources, not a proven research gap.

5. Questions, hypotheses and objectives

Main question: to what extent can the prototype support decisions that improve the cost balance? Objectives: audit data, compare predictors, validate prospectively and evaluate authorized trials. Proposed predictive hypothesis: a selected predictor reduces P80 error against baselines under an identical evaluation protocol. Economic effects require separate evaluation without promised percentages.

6. Methodology

Design science with quantitative evaluation; 120 historical events reported and approximately 180 future events projected. Confirm whether an event represents a ring, blast or stope; separate time/groups, fit preprocessing/tuning on training only, and evaluate new data before updating. Measure MAE/RMSE, compatible costs and production separately. An uncontrolled before/after comparison does not establish causality.

7. Ethics and permissions

Clarify academic analysis, publication and field permissions; the reported approving authority is General Management. Confirm university ethics-review requirements; no approval or unrestricted publication right is assumed.

8. Expected outputs

An evaluated prototype, reproducible protocol and comparisons, including negative findings. No savings, ANN superiority or production improvement is guaranteed.

9. Timeline and research budget

Relative plan: M1 permissions/audit; M1-M6 observation; M2-M3 development; M4-M6 prospective assessment and authorized trials; final analysis afterward. Calendar dates and cost estimates remain pending; the operational monthly budget is not the research budget.

10. Bibliography

Hevner et al. (2004), Shields et al. (2025), Amoako et al. (2022) and Manzoor (2023); see Session 4 references.